Lab 2

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1. Introduction

1.1. Purpose

This document provides the software requirements specification for Gold Star, a web-based productivity platform.

1.2. Scope

Individuals with ADHD diagnoses or ADHD-like traits often struggle with consistent routine development due to various productivity-altering traits. Routines last for short time periods, and are vulnerable to persistent Routine Adaptations leading to frustration and reduced self-efficacy. Gold Star is a web-based habit-building and time management platform to help users affected by ADHD. Gold Star helps organize schedules, break down tasks, and build sustainable routines. Gold Star intends to make daily routine development easier to start, maintain, and complete.

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1.3. Definitions, Acronyms, and Abbreviations

Activity Alternation – Switching from an activity requiring high mental focus to an activity requiring low mental focus.

Activity Switching – The cognitive process of disengaging from one task and shifting mental focus to another task.

Adaptive Scheduling – A schedule readjustment based on the Scheduler's selected productivity levels, task prioritizations, and mood to maintain flexibility.

Application Programming Interface (API) - A set of standard rules, protocols, and definitions allowing different software applications to interact with one another.

Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.

Calendar View – Gold Star's calendar interface mechanic for Schedulers to view all tasks planned in a given month or day.

Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.

Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.

GitHub - A cloud-based platform where developers can collaborate on software code.

Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD.

Graphical User Interface (GUI) – A visual interface intended for users to manipulate data or view output through interactive visual components rather than text commands. Examples: buttons, windows, or icons.

Habit Stack – A smaller isolated sequence of tasks from a Scheduler's larger schedule. Habit Stacking allows the Habit Stack to resize as Schedulers' habits change.

Habit Stacking – Interspersing new habits with existing habits to make task completion easier without feeling overwhelmed.

Hyperactivity - A state of being unusually or abnormally active.

HyperText Markup Language (HTML) - The standard markup language used to create and structure webpages.

JavaScript (JS) - A lightweight programming language used to build dynamic interactivity on web pages.

Jest - An open-source JavaScript testing framework.

JSDoc - An API documentation generator and syntax standard for JavaScript.

MySQL – An open-source RDBMS that stores, manages, and retrieves data for Gold Star's prototype.

Node.js – An open-source JavaScript runtime environment to execute JavaScript code outside of browsers for functionalities like server management.

Non-Rigid Scheduling – To schedule a task which can be completed anytime within a set timeframe. Example: a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.

Non-Rigid Task – Tasks lacking required starting times and deadlines. Non-Rigid Tasks require an estimated timeframe for completion. Non-Rigid Tasks are scheduled by Adaptive Scheduling.

Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute breaks. The standard Pomodoro process repeats this technique 4 times, then a longer break between 20 to 30 minutes is taken before the process repeats.

PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data.

Reinforce - To make something stronger, more effective, or more compelling by adding support.

Relational Database Management System (RBMS) - Software used to store, manage, query, and retrieve data structured in table form.

Rigid Scheduling – To schedule a high-priority task which must either be initiated at or completed before an established time. Example: a task which must start at 6:00 P.M. or must be completed by a deadline of 6:00 P.M.

Rigid Task – Non-negotiable tasks requiring a specific starting time or deadline for completion. Rigid Tasks are scheduled by Rigid Scheduling.

Routine Adaption – A requirement for an individual to change one's schedule or create new routines.

Routine - A sequence of actions regularly followed, or a fixed program.

Scheduler – A Gold Star user. Schedulers are responsible for adding tasks to their schedule, editing existing tasks within their schedules, and reviewing their existing schedules.

Task Deviation – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.

Task Management – The process of creating a new task, or modifying an existing task.

Visual Studio Code (VSCode) - A development environment used for software engineering and web development.

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1.5. Overview

Section 2 contains a summarized description of Gold Star.

Section 3 contains each requirement, organized by feature pertaining to Gold Star's intended prototype.

2. Overall Description

2.1. Product Perspective

Gold Star is a web-based habit-building platform for individuals with ADHD who struggle with routine development. Gold Star offers a low-stakes environment for organizing and monitoring daily tasks, schedules, and habits without feeling overwhelmed. Gold Star offers support for easier routine development and preserving long-term goal consistency.

2.2. Product Functions

Gold Star's major functions intend to improve productivity and habit-building. The user interacts with Gold Star through a web Graphical User Interface centered on the Calendar View interface. Calendar View serves as Gold Star's primary calendar interface for maintaining daily schedules. Task Management supports adding new tasks to a specified day, and modifying existing tasks. Habit Stacking lets users associate new tasks with existing routines. Adaptive Scheduling allows schedule readjustment based on changes in task priorities and daily productivity. Schedules are also readjusted based on users' moods specified by the Daily Mood Check-In function. Task Prioritization enables users to decide which tasks are more important to complete.

2.3. User Characteristics

The "Scheduler" is Gold Star's only user role. Schedulers rely on Gold Star to maintain daily responsibilities, reduce daily planning difficulties, and break down tasks. Schedulers are responsible for adding tasks, modifying existing tasks to adapt to changing priorities, and verifying their finalized schedules are realistic and manageable.

2.4. Constraints

N/A

2.5. Assumptions and Dependencies

N/A

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